

Reviewer Pack — Minimal Rank of Symplectic Leaves vs Communication Complexit...

Entry 5247e427051e · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Symplectic Leaves vs Communication Complexity for k-CNF

Entry ID: 5247e427051e **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-23 21:53:54 UTC

1. Conjecture

Field A (mathematical branch): Symplectic Geometry **Field B** (complexity object): Communication Complexity

Statement:

{‘quantitative relation’: ‘ $E[\psi(S)] = \Theta(\log(n))$ for all symplectic leaves S such that there exists a protocol with communication complexity $\leq c \cdot n$ for k -CNF satisfiability, where $\psi(S)$ is the minimal rank of the corresponding symplectic leaf.’, ‘falsifier’: ‘There exists a symplectic leaf S with $\psi(S) = o(\log(n))$ and a k -CNF satisfiability protocol with communication complexity $\leq c \cdot n$.’}

Rationale (proposer’s reasoning):

{‘explanation’: ‘Symplectic geometry provides a geometric framework that could potentially encode the structure of computational tasks. If certain symplectic leaves can be shown to have a low rank while corresponding protocols for NP-complete problems like k -CNF satisfiability require significant communication, it would suggest a deep connection between symplectic geometry and complexity theory.’, ‘potential explanation’: ‘Symplectic leaves could serve as geometric representations of computational processes, allowing us to leverage geometric properties to understand complexity bounds.’}

Taxonomy category: AC0_PARITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 0c5697e3f3132240

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all k -CNF instances with $n \leq 40$ variables and 30 random seeds, the average minimal rank of the corresponding symplectic leaves satisfies $E[\psi(S)] = \Theta(\log(n))$, where $\psi(S)$ is the minimal rank of the symplectic leaf. The conjecture is falsified if any seed yields a symplectic leaf with $\psi(S) < \log(n)$ or an average rank $> 3 \cdot \log(n)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.3s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_k_cnf(n):
        clauses = []
        for _ in range(2**n):
            clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
            if any(clause[i] == -clause[j] for i in range(n) for j in range(i+1, n)):
                continue
            clauses.append(clause)
        return clauses

    def symplectic_rank(matrix):
        n = len(matrix)
        rank = 0
        for i in range(n):
            if matrix[i][i] != 0:
                rank += 1
                for j in range(i+1, n):
                    factor = matrix[j][i] / matrix[i][i]
                    for k in range(n):
                        matrix[j][k] -= factor * matrix[i][k]
        return rank

    def calculate_minimal_rank(clauses):
        n = len(clauses[0])
        symplectic_matrix = [[0] * n for _ in range(n)]
        for clause in clauses:
            for i, lit in enumerate(clause):
                if lit > 0:
```

```

        symplectic_matrix[i][lit-1] += 1
    else:
        symplectic_matrix[lit-1][i] += 1
    return symplectic_rank(symplectic_matrix)

n = random.randint(5, 40)
clauses = generate_k_cnf(n)
minimal_rank = calculate_minimal_rank(clauses)

return {
    "metric_name": "Minimal Rank",
    "metric_value": minimal_rank,
    "instances_tested": 1,
    "conjecture_holds": minimal_rank >= math.log(n),
    "counterexample": ""
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"minimal_rank_too_low\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means we cannot confirm or refute the conjecture based on the pre-registered support condition. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	10252
2	propose	ollama_remote	glm4:latest	0	0	13239
3	preregistration	ollama_remote	glm4:latest	0	0	5969
4	novelty	ollama_remote	glm4:latest	0	0	5827
5	novelty	ollama_remote	glm4:latest	0	0	5261
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14286
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10235
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15272
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9087
10	judge	ollama_remote	glm4:latest	0	0	37308

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 126734 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in *research/audit/5247e427051e.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/5247e427051e.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/5247e427051e.tar.gz* (if generated)